

Allison Zhao

 [allison-stack](#)  [Allison Zhao](#)  allisonzhao.uni@gmail.com  289-775-9755  [allisonzhao.vercel.app](#)

EDUCATION

McMaster University

Sept. 2022 – May 2027

Bachelor of Applied Science in Computer Science with Co-op

4.0/4.0 GPA

- 2022-2025: Deans' Honour List
- 2025: The Dr. Harry Lyman Hooker Scholarship
- 2023: The Pollock Family Academic Grant
- 2022: The McMaster University Award of Excellence
- **Extracurriculars:** DeltaHacks VP Logistics, McMaster Start Coding Mentor

SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript, C/C++, HTML, CSS, Haskell, Elm

Frameworks and Libraries: NextJS, React, Tailwind CSS, Flask, Spring Boot

Tools: Git/GitHub, Linux, Bash Scripting, PostgreSQL, MySQL, Google Cloud Platform, Docker

EXPERIENCE

DevOps Engineer

May 2025 – Present

Huawei Technologies Canada

- Automated TPC-H power/throughput benchmark execution for GaussDB, a cloud-native distributed relational database, by developing Bash scripts to compile per-query QphH scores and timing data; contributed to a 30TB **world record** of 39.974M QphH (+43% over prior record) and 10TB record of 16.2M QphH using half the compute of the prior record holder
- Identified **6 pre-release bugs** in TaurusDB Serverless's online-resizable buffer pool by designing a functional verification test plan covering concurrent resize, in-progress checkpoints, dirty pages, vacuum, and master/standby replication configs
- Restored regression test coverage for TaurusDB for PostgreSQL's Storage Abstraction Layer by debugging and remediating **30+** failing test cases — fixing broken logic, porting MySQL-specific tests to PostgreSQL compatibility, and resolving codebase drift since 2018; tests integrated into the development team's CI pipeline

Undergraduate Research Assistant

May 2024 – Aug. 2024

McMaster University

- Implemented **168 RISC-V instructions** across 4 ISA extensions (RV32/64I, RV32/64M, RV32F, RV64D) in Haskell by translating official ISA specifications into formal type-class semantic models covering both general-purpose and floating-point register files
- Achieved **100% ISA compliance** by building a Dockerized QEMU test harness of 82 C programs that cross-validated instruction output against the Haskell interpreter, resolving correctness bugs in signed/unsigned overflow, division-by-zero, and floating-point bit-exact transfer edge cases
- Implemented 3 vectorized **IBM MASS** math library functions (logarithm, power, cube root) targeting POWER ISA SIMD vector instructions, applying Horner's method polynomial approximation and fused multiply-add error compensation to achieve IEEE 754-accurate single-precision results

PROJECTS

ASCII-Battle-Game | Python, TCP Sockets

Apr. 2025

- Built a networked multiplayer battle game supporting 4 concurrent players by implementing an object-oriented TCP socket server with real-time game state synchronization and a custom application-layer protocol for commands (move, attack, fireball, chat)

mac-FAQ-chatbot | Python, Google Cloud Functions, BigQuery, Dialogflow

Apr. 2024

- Wrote BigQuery SQL queries using UNNEST to filter FAQ responses across array-typed topic columns, and built Cloud Functions webhook endpoints to route Dialogflow intents to the appropriate query and return contextual answers to McMaster students via a serverless API